

Industry Preparation and Next Steps

Week 20

- Transition from course to industry or further study
- Explore career opportunities and pathways
- Reflect on your learning journey

Learning Objectives

- Identify key industry pathways for AI and ML roles
- Recognise transferable skills developed during the course
- Prepare for entry into the workforce or further education
- Create a personal action plan for your next steps

Agenda

- Industry pathways and workplace applications
- Reviewing course skills; mapping to job roles
- Career and study options after the course
- Portfolio and self-assessment
- Activities and next steps

The AI and ML Industry Landscape

- Rapidly growing need for machine learning and AI skills in business and government sectors
- Typical roles include ML Engineer; AI Developer; Cloud Data Specialist; MLOps Technician
- Employers often look for practical experience and portfolio evidence

Transferable Industry Skills

- Programming; scripting; and automation using Python and Bash
- Cloud configuration and deployment using Azure and Linux environments
- Building and training machine learning models using PyTorch
- Parallel and distributed computing skills for scalable solutions
- Following responsible AI standards; MLOps; and model documentation

Mapping Course Skills to Job Tasks

- Writing scripts for ML workflows; automating routine processes
- Configuring GPU VMs and managing cloud resources
- Running experiments; monitoring models; using experiment tracking tools
- Building and deploying production-ready AI models
- Documenting code and pipelines for workplace handover

Example Job Roles and Descriptions

- Junior ML Engineer. Implements ML workflows; maintains models and scripts
- Cloud AI Support Technician. Configures compute environments; troubleshoots deployments
- AI Developer. Designs; tests; and deploys neural network solutions
- MLOps Assistant. Automates model training and deployment; maintains experiment records

Where Can These Skills Take You?

- Employment opportunities in software companies; government; health tech; mining; and more
- Roles ranging from technical support; development; analysis; to operations management
- Strong foundation for career progression in AI and cloud engineering fields

Further Study Pathways

- Diploma or Advanced Diploma in Software Development; Cloud Engineering; or AI (TAFE)
- Bachelor of Computer Science; Data Science; or AI (university)
- Industry certificates: Microsoft Azure; AWS Certified ML; Certified MLOps Engineer

Reviewing Your Portfolio

- Ensure you have clear documentation for all major projects and assessments
- Showcase end-to-end ability. Include code samples; experiment reports; cloud deployment cases
- Reflect on growth: What skills have you developed since Week 1?

Case Study: AI in Practice

- Example: Local government using GPU-accelerated computing for infrastructure planning
- Team built a scalable pipeline using PyTorch and cloud infrastructure
- Success based on ability to work across scripting, modelling, and deployment

Activity: Job Search and Skills Mapping

- Find 2-3 relevant job ads in AI, ML, or cloud roles
- Identify which course skills are requested by employers
- List any additional skills or certificates to consider for the future

Activity: Personal Action Plan

- Outline your goals for the next six months after this course
- List your target roles or study pathways
- Note immediate next steps (e.g., updating resume, applying for jobs, contacting potential mentors)

Preparing Your Final Portfolio

- Double-check that all assessments are complete and documented
- Follow documentation standards from Week 18; include experiment tracking and cloud setup info
- Submit your portfolio by the due date for final review

Assessment & Reassessment

- All assessment tasks must meet competency standard to pass
- If reassessment required, use Weeks 19-20 for support and revision
- Reach out to your lecturer for feedback and help

Reflection: Looking Back and Ahead

- Think about your progress from scripting basics to distributed ML
- Acknowledge your persistence and commitment
- Prepare to showcase your skills and keep learning

Final Thoughts

- The tech industry values practical skills, curiosity, and adaptability
- Keep building your portfolio and learning through real-world projects
- You are ready for the next stage. Good luck in your AI and ML journey!